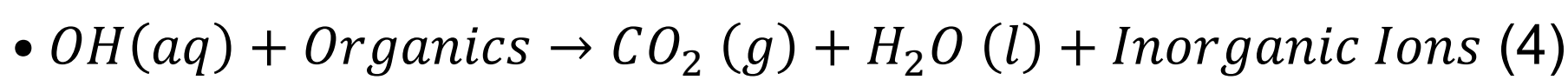
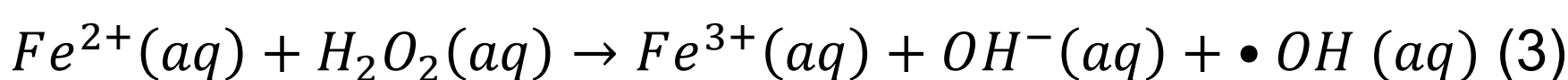
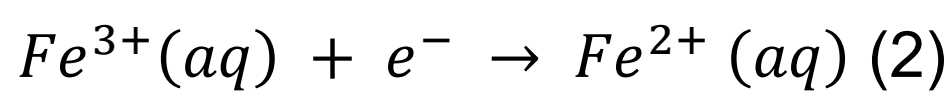
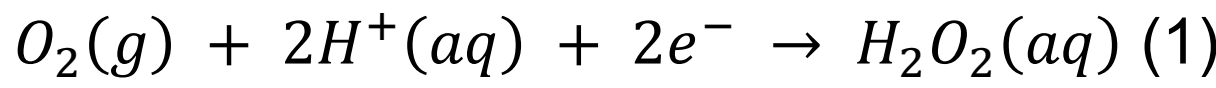


I. Background

The Indonesian textile sector contributes roughly IDR 218.2 trillion to national GDP (2024), yet its wastewater carries toxic, carcinogenic synthetic dyes such as methylene blue. Permen LHK No. 16/2019 limits effluent color to 200 Pt-Co, while industrial wastewater has been measured at up to 1073.47 Pt-Co, meaning **at least 82% degradation is required** before legal discharge. A prior UV-based adsorption-photo-Fenton reactor (BRIN & UNS) reached only 50% degradation and imposed UV-related safety, observability, and space constraints.

Effiton removes the UV dependency through the electro-Fenton (EF) route, generating H_2O_2 electrochemically and continuously regenerating the Fe^{2+} catalyst:



The resulting $\bullet OH$ radical is among the strongest known oxidants, breaking down persistent dye molecules entirely UV-free.

II. System Description

Effiton is organized around eight functional blocks: a 250 mL reactor core with electrodes, heater, stirrer, and gas/pH/temperature sensors; a multi-rail power supply (83/12/6/5 V); a 7-inch touchscreen interface; seven peristaltic pumps; reagent containers; a custom spectrophotometer; an ESP32 controller; and a waste container. Running FreeRTOS, the ESP32 coordinates 14 concurrent tasks through five states (Input Initialization $\rightarrow$ Initialization $\rightarrow$ Idle $\rightarrow$ Execution $\rightarrow$ Flushing), enabling scheduled adsorption-then-electro-Fenton runs with per-5-minutes spectral adsorption measurement and automatic flushing.

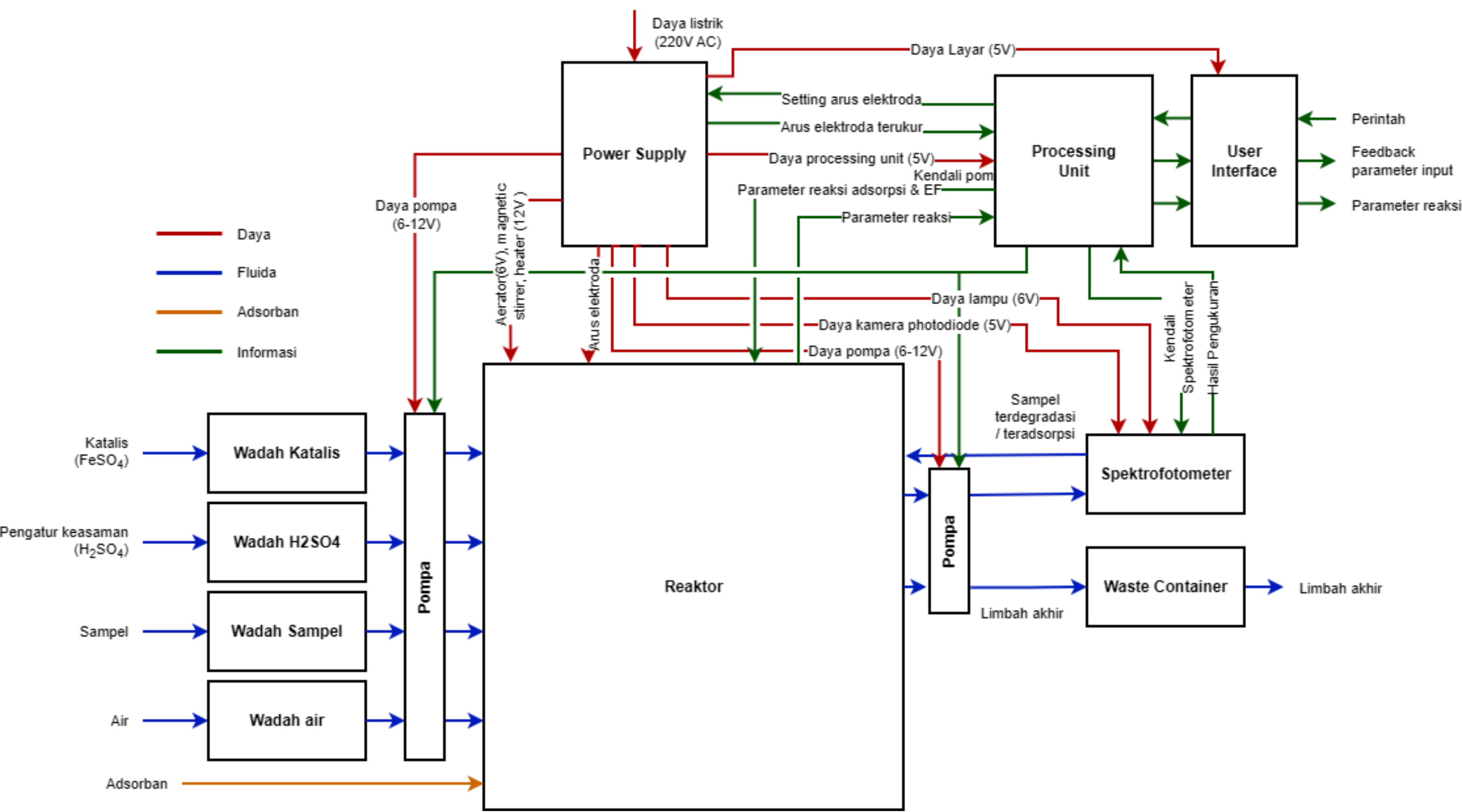


Fig. 1. Effiton main system block diagram : power, material, and information flows.

III. Hardware Design

A. Reactor System

A PI-Controlled magnetic stirrer keeps the zeolite suspended and mixes the 250 mL EF cell, while a PID-controlled heater holds 25-40 °C via a DS18B20 sensor and an SSR-switched heater. Gas, pH, and temperature sensors on the reactor lid enable live monitoring (Fig. 3).

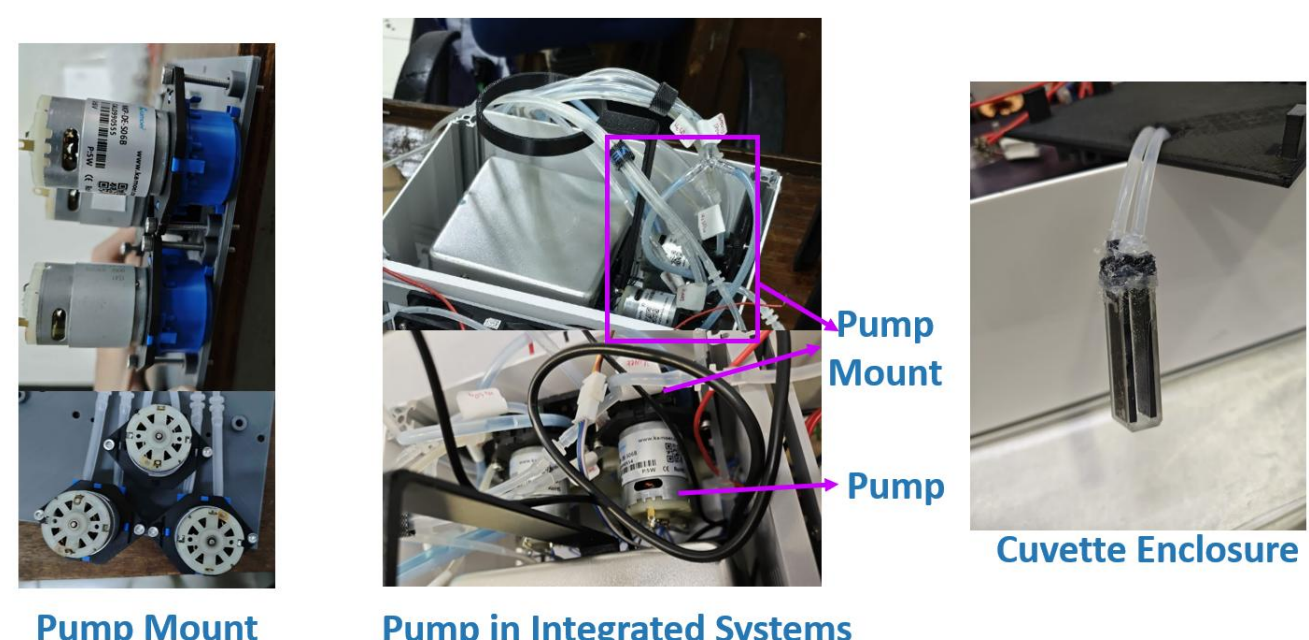


Fig 4. Reagent Dosing Hardware

C. Electrode Current Source

A BJT voltage-controlled current source (paralleled MJ15024 transistors, LM358 feedback) converts the ESP32 DAC output (0-3.3V) into electrode current with $R^2 = 1.000$ across the full 0–220 mA design range and 0.003 V_{rms} ripple (Fig. 5):

$$I = 0.063 \cdot V_{in} + 0.00533 \text{ A} \quad (5)$$

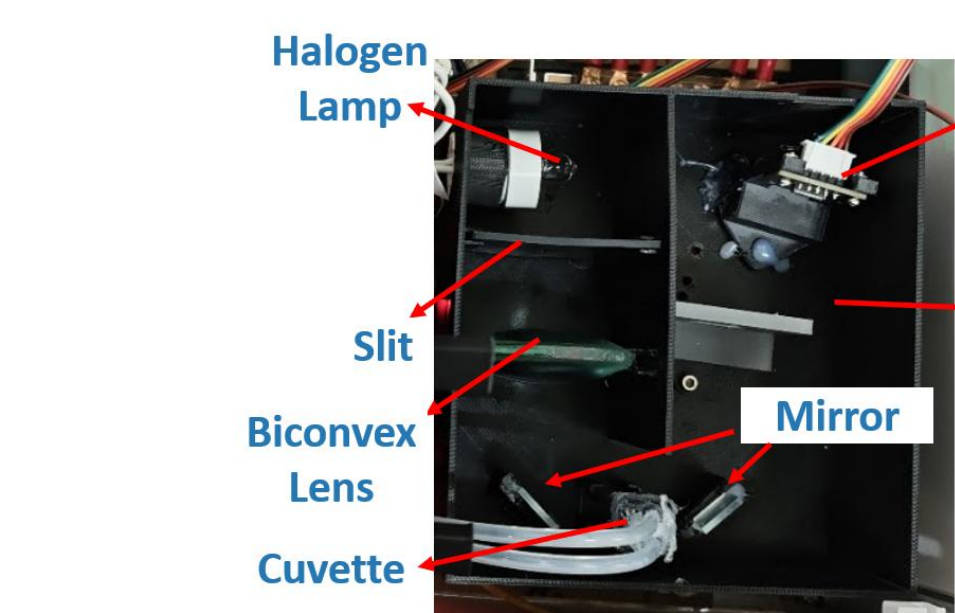


Fig 5. Spectrophotometer

E. User Interface & Controller

A 7-inch touchscreen with keypad fallback exposes scheduling, live sensor monitoring, and spectrum visualization through a guided menu, letting users adjust stirrer speed, temperature, electrode current, and reaction duration, or trigger the flushing sequence manually (Fig. 7).

B. Reagent Dosing

Two Kamoer NKP-DE-S10D pumps dosed $FeSO_4$ / H_2SO_4 at a calibrated 30 mL/min to hold pH 2-4 for efficient $\bullet OH$ generation, as well as spectrophotometer circulation; 120 mL/min transfer pumps handle sample loading and end-of-cycle fresh-water flushing (Fig. 4).

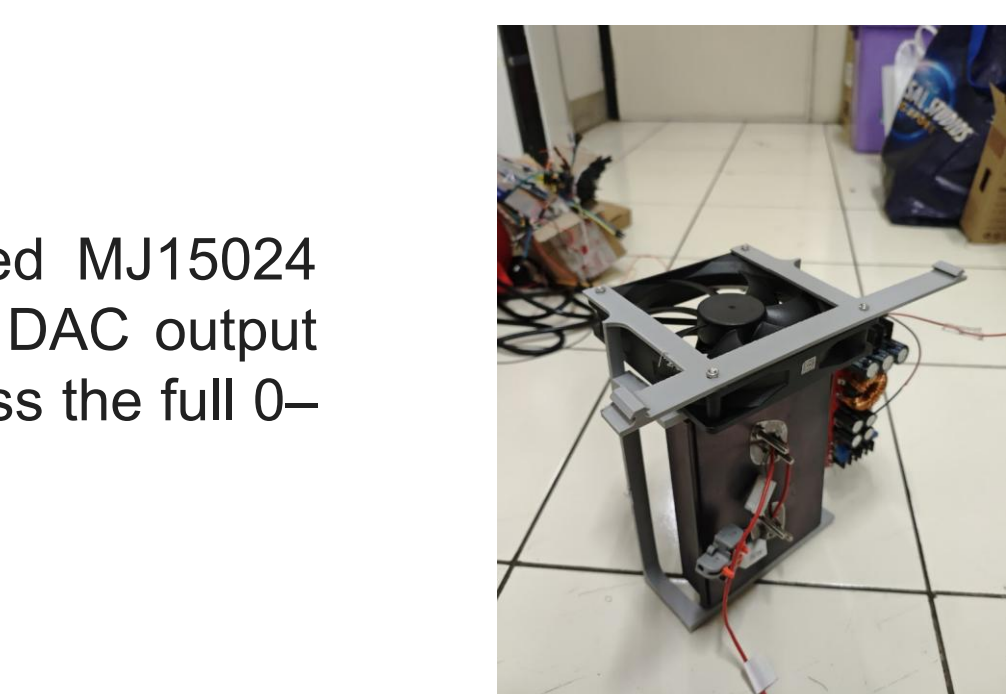


Fig 5. Current Source Hardware

D. Spectrophotometer

A custom diffraction-grating spectrophotometer (600 lines/mm grating, TSL1401CL 128-pixel linear sensor, 3.13 nm/pixel over 400–800 nm) scans reactor fluid drawn through the cuvette line; wavelength accuracy was verified at PPNN ITB with 1.67% error for methylene blue (Fig. 6).

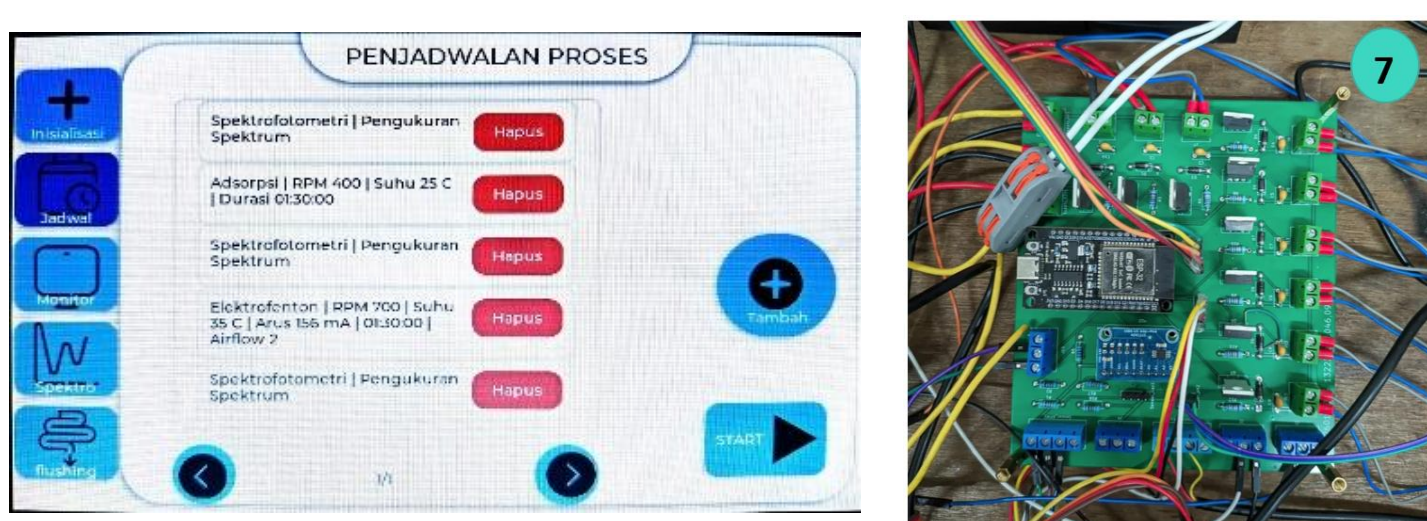


Fig 5. Scheduling in UI and Controller PCB

IV. Result



Fig. 2. Physical implementation of Effiton.

What can Effiton do?

Effiton runs the entire treatment cycle by itself: schedule a run from the touchscreen, and the system doses the reagents, maintains the reaction conditions, measures the color change every five minutes, and flushes itself clean when finished, no manual sampling, no external lab instruments, and no UV radiation. Anyone with basic laboratory training can operate it directly on a benchtop.

One Integrated benchtop unit, from raw dye sample to treated water

Fully automatic operation at just 192 W(Max), about the power of two laptops

Product Characteristics

- Automated & scheduled:** adsorption, electro-Fenton reaction, measurement, and flushing run on a user-predefined schedule
- Adjustable:** stirrer speed, temperature, electrode current, and duration are all tunable from the touchscreen
- Self-monitoring:** a built-in spectrophotometer tracks degradation periodically, so results are visible as they happen
- Compact & efficient:** 50×25×15 cm and energy-efficient, fits directly on a laboratory bench
- Safe:** completely UV-free, replacing the hazardous UV lamps of previous reactor designs

82.6% degradation in 60 min

≈ 96% in 120 min, by electro-Fenton run at 126 mA

Operated at 126 mA, methylene blue absorbance at $\lambda_{peak} = 721.3$ nm fell from 0.46 a.u. to 0.08 a.u. within 60 minutes, an 82.6% degradation that meets the $\geq 82\%$ regulatory target within one hour and reached 0.02 a.u. (≈ 96%) after 120 minutes. This far exceeds the 50% of the prior UV-based system while eliminating its safety constraints. The onboard spectrophotometer captured full 400–800 nm spectra every 5 minutes throughout the run.

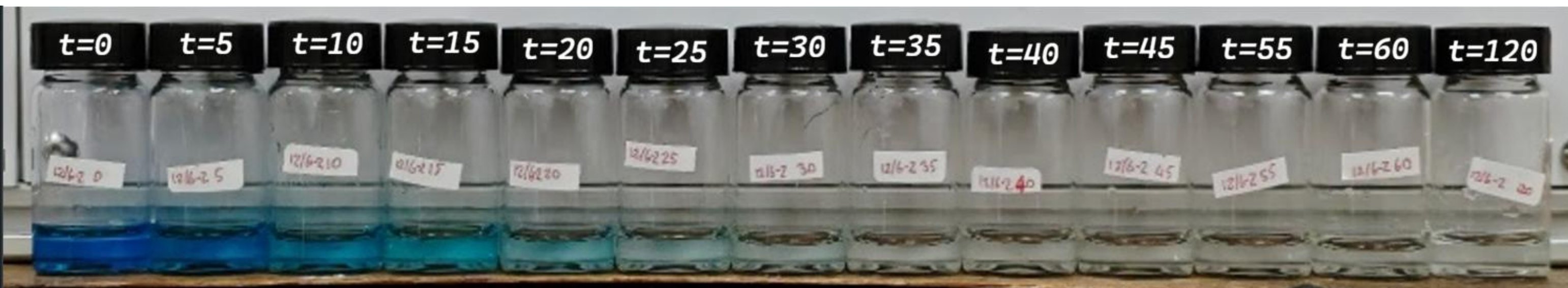


Fig. 8. Methylene blue samples over the 120 minute EF run.

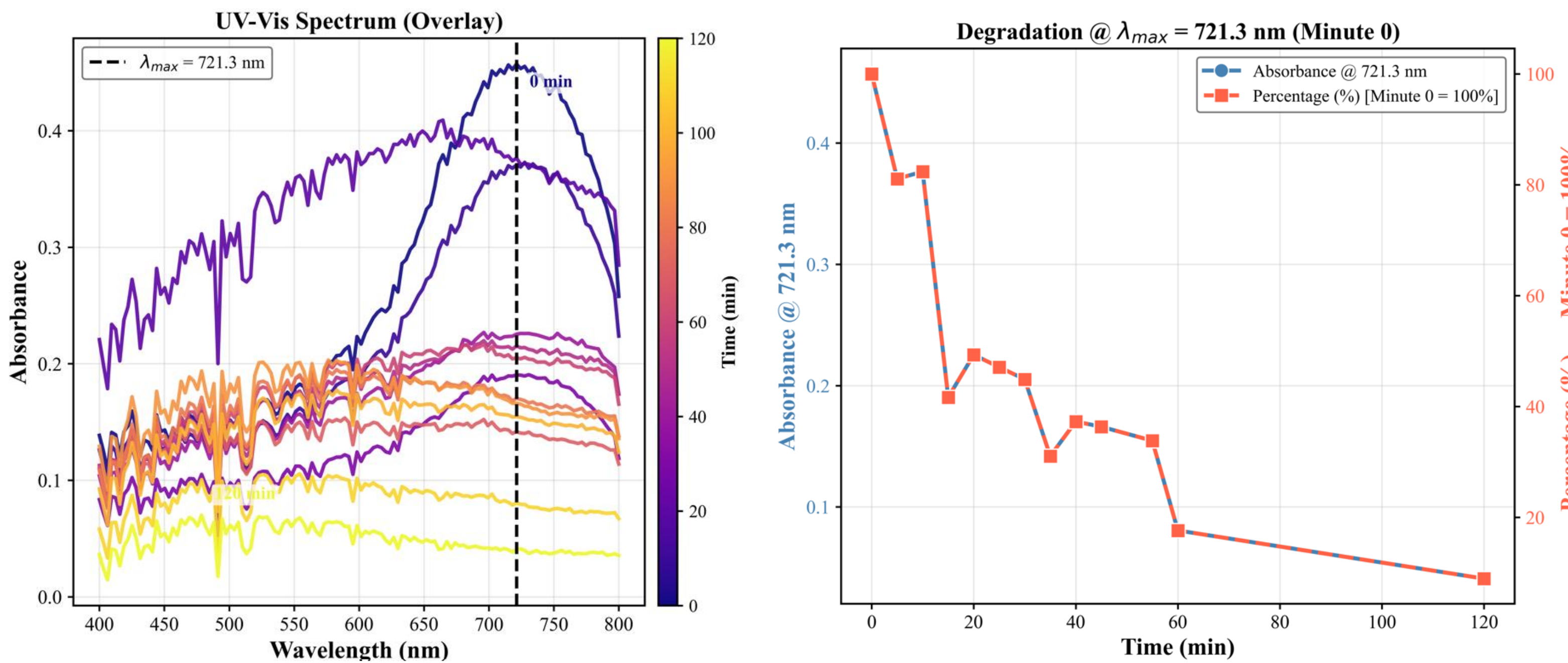


Fig. 9. UV-Vis spectra for every sampling point, 400-800 nm.

Fig. 10. Degradation over time at $\lambda = 721.3$ nm.

V. Conclusion

Effiton unifies zeolite adsorption, homogeneous electro-Fenton degradation, and real-time spectrophotometric monitoring into a single automated, UV-free benchtop platform that meets Indonesia's textile discharge requirement: 82.6% methylene blue degradation within one hour (≈ 96% in two) at 126 mA, with 9 of 10 product characteristics and all 18 subsystem specifications met. Future work covers validating the combined adsorption + Electro Photo Fenton sequence, spectrometer mirror stabilization, low-RPM stirrer sensing, and Y-junction management.

[1] R. N. Thahani, A. Novel, and R. Javrico, "Design and Implementation of Filter Reactor Using Adsorption and Fenton Reactor for Textile Wastewater."

[2] A. Kuleyin, A. Gök, and F. Akbal, "Treatment of textile industry wastewater by electro-Fenton process using graphite electrodes in batch and continuous mode," J. Environ. Chem. Eng., vol. 9, no. 1, p. 104782, 2021.

[3] H. He and Z. Zhou, "Electro-Fenton process for water and wastewater treatment," Crit. Rev. Environ. Sci. Technol., vol. 47, no. 21, pp. 2100–2131, 2017.

[4] Kementerian LHK, Peraturan Menteri LHK No. 16/2019 (effluent color limit 200 Pt-Co).

[5] A. Maulana Hardi, "Pengolahan air limbah industri tekstil dengan metode fotokimia uv-h202," Universitas Indonesia Library.